Sujith Makam

Software Engineer

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Leet Code

Education

Indian Institute of Information Technology (IIIT), Dharwad **DEC 2021 - JULY 2025**
B.Tech in Computer Science and Engineering *CGPA: 8.41*

Technical Skills

Languages: C, C++, Python, JavaScript, TypeScript
AI/ML: Generative AI, Agentic AI, LLMs
Cloud & DevOps: Azure (Azure AI, Azure SQL, ADLS Gen2, Azure Functions, Event Hub, IoT Hub), AWS(CDK, Lambda, EventBridge, SSM, boto3, Amazon Connect), Docker
Data & Analytics: Microsoft Fabric (Lakehouse, Pipelines, Eventstream, KQL, Real-Time Intelligence), Power BI, Apache Spark, Delta Lake, Microsoft Purview
Backend:NodeJs, ExpressJs, REST APIs
Frontend:ReactJs
Databases:MongoDB, MySQL
Tools: Git, Cognigy, Genesys, Microsoft Fabric.

Experience

CGI Jul. 2025 - Present

Software Engineer

- Built a unified multi-source data ingestion platform on Microsoft Fabric, consolidating streaming and batch data from Azure SQL, AWS S3, ADLS Gen2, Azure IoT Hub, and Azure Event Hubs into a governed Medallion Lakehouse (Bronze/Silver/Gold), enabling real-time operational analytics and Power BI reporting for enterprise contact center insights.
- Implemented Microsoft Purview data governance across the Fabric platform, enforcing data lineage tracking, access controls, and compliance policies over all ingested data assets.
- Engineered end-to-end **Infrastructure-as-Code (IaC) using AWS CDK (TypeScript)** to provision and deploy complete Amazon Connect environments, **reducing manual setup time by 80 %** and enabling scalable, automated cloud contact-center deployments across regions.
- Built production-grade **Amazon Connect Disaster Recovery (DR) synchronization solution** using boto3, AWS SDK, Lambda, EventBridge, and SSM; automated cross-region configuration sync to **maintain DR as a near-real-time mirror of production, achieving 100 %** compliance and zero-downtime failover readiness.
- Designed and deployed an **Agentic AI-driven orchestration framework integrating Azure AI, Cognigy, Genesys, and AWS services**, automating first-level customer interactions and **reducing manual agent escalations by 85 %** while improving decision accuracy and process reliability in enterprise workflows.

Indian Institute of Science Bangalore May. 2024 - July. 2024

Research Intern

- Investigated LLM pruning techniques to improve GPT-2 efficiency, enabling optimized attention and feed-forward layer execution. **Achieved a 5.2% reduction** in instruction cache misses through tiled matrix multiplication implementation.
- Conducted detailed profiling of GPT-2 using Intel VTune, identifying bottlenecks and enhancing cache efficiency, which resulted in a 4x speedup in attention layer execution.
- Optimized bottlenecks in GPT-2 attention and feed-forward layers, reducing memory bandwidth usage by 34.8% and improving compute efficiency.

Projects

Enterprise Multi-Source Data Platform on Microsoft Fabric

- Architected a production-grade Medallion Lakehouse on Microsoft Fabric, ingesting data from 5 heterogeneous sources — Azure SQL DB, AWS S3, ADLS Gen2, Azure IoT Hub, and Azure Event Hubs — using Fabric Pipelines (batch), Eventstream (real-time), and OneLake Shortcuts for zero-copy external data access.
- Implemented Medalion Architecture (Bronze/Silver/Gold) transformation layers using Apache Spark notebooks in Fabric; handled schema evolution, incremental updates, and late-arriving data with configurable retry mechanisms and idempotent processing logic.
- Enforced enterprise data governance using Microsoft Purview — catalogued all data assets, a• Delivered interactive Power BI reports on the Gold layer, enabling business stakeholders to consume unified, governed, and always-fresh analytical data across batch and streaming workloads from a single semantic model.pplied sensitivity labels, tracked end-to-end lineage from ingestion to consumption, and configured role-based access controls across workspaces.
- Built real-time KQL-based dashboards using Fabric Real-Time Intelligence and automated security alert pipelines to detect anomalous event patterns from IoT Hub streams with sub-second latency.
- Delivered interactive Power BI reports on the Gold layer, enabling business stakeholders to consume unified, governed, and always-fresh analytical data across batch and streaming workloads from a single semantic model.

Lightweight IoT Intrusion Detection System (Edge-AI)

- Developed a scalable Curriculum Learning framework with XAI (LIME) validation, utilizing optimized GRU/LSTM attention layers to progressively detect complex cyber-attacks.
- Attained 98 % accuracy on the CIC-IoV-2024 dataset while reducing model size to 367KB via quantization and pruning, enabling high-efficiency deployment on resource-constrained edge devices.

FAPL-DM-BC: Blockchain-Federated Security Framework

- Designed a privacy-preserving IoV framework merging Federated Learning with Blockchain smart contracts for decentralized model validation and tamper-proof provenance.
- Achieved 93.1% accuracy with 42ms latency by engineering a Dockerized microservices architecture (using SMPC) and integrating a dual-model XAI feedback loop (SHAP/LIME) for real-time threat detection.

Research Publications

IEEE COMSNETS 2026

"FAPL-DM-BC: Adaptive Privacy-Aware, Blockchain-Backed FL Framework with Dual-Model XAI for the IoVs"

IEEE FNWF 2025

"Enhancing IoT Network Security through Adaptive Curriculum Learning and XAI"

IEEE FNWF 2025

"FedGuard-IIoT: A Scalable Federated Learning Architecture for Network-IDS in the Industrial IoT"

IEEE SPACE 2025

"Predicting Stellar Metallicity: A Comparative Analysis of Regression Models for Solar Twin Stars"

Achievements

TCS TechBytes: Dharwad Regional Finals Runner Up

A Prestigious Tech Quiz conducted by TCS and BITES in Katakana

State Rank Holder: Andhra Pradesh State Board Topper

Scored 990/1000 in my state board examinations